

1. Project Overview

We are making a project to convert text (Document) to speech (Audio). The document includes TXT, PDF, newsletters, or others. We implement AWS tools: Amazon Polly, Amazon S3 and Amazon Lambda

2. Services Used:

Amazon Polly: Converts text to speech with various features.

AWS IAM: Ensures secure access by managing user permissions.

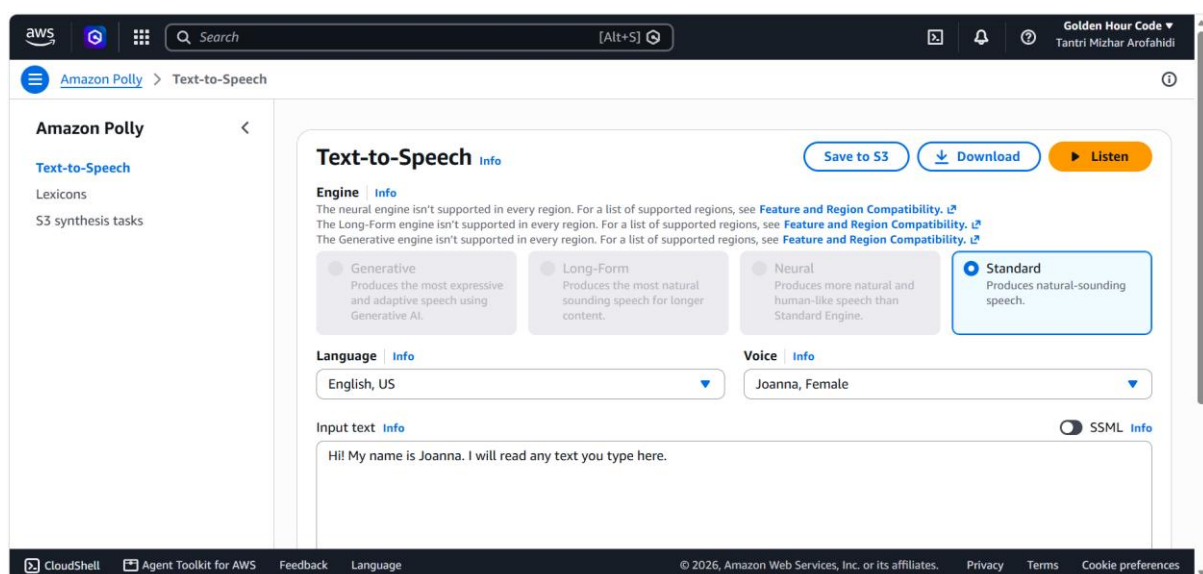
AWS S3: Stores configuration name files (txt and audio).

AWS Lambda: Connection txt file to audio.

3. Estimated Cost:

Cost: Free (When using the AWS Free Tier)

4. Amazon Polly

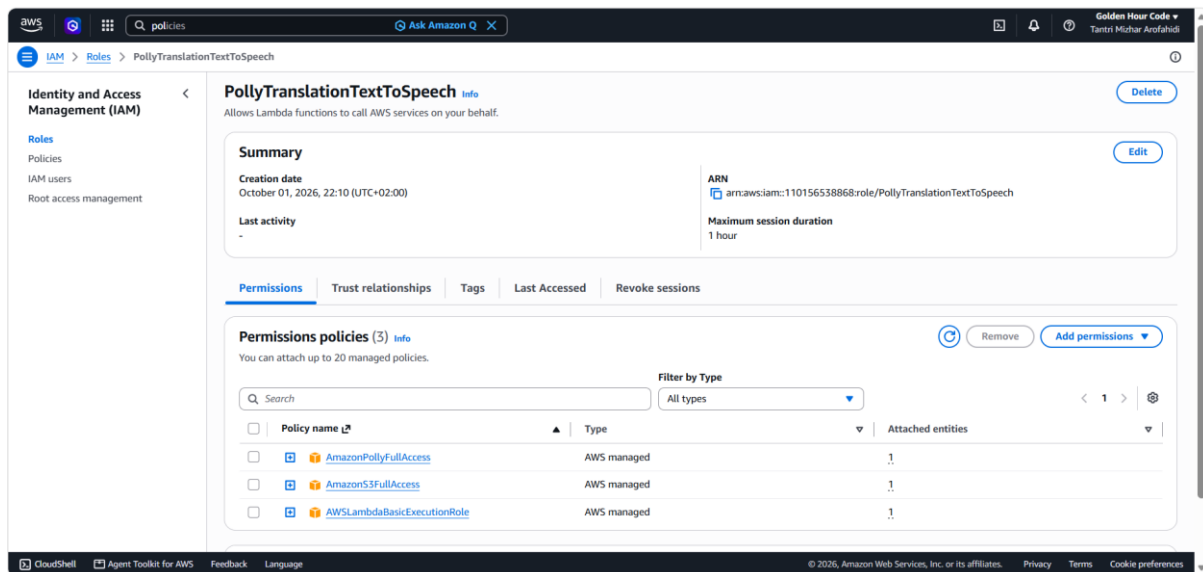


5. IAM Role

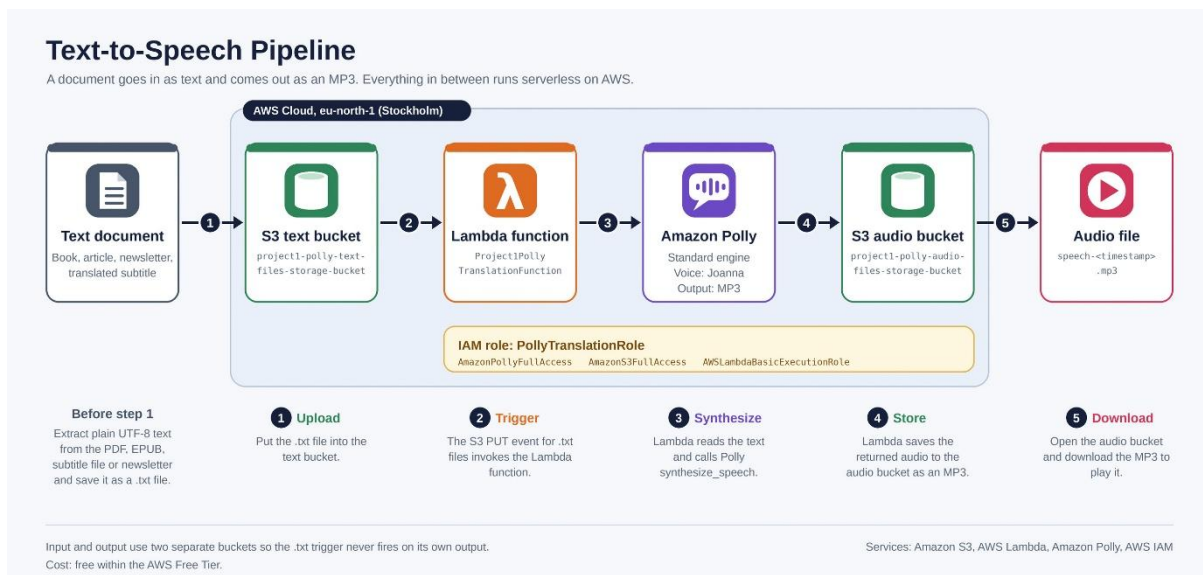
We are creating name role in IAM:

RoleName: PollyTranslationRole

Policies: AmazonPollyFullAccess, AmazonS3FullAccess, and AWSLambdaBasicExecutionRole



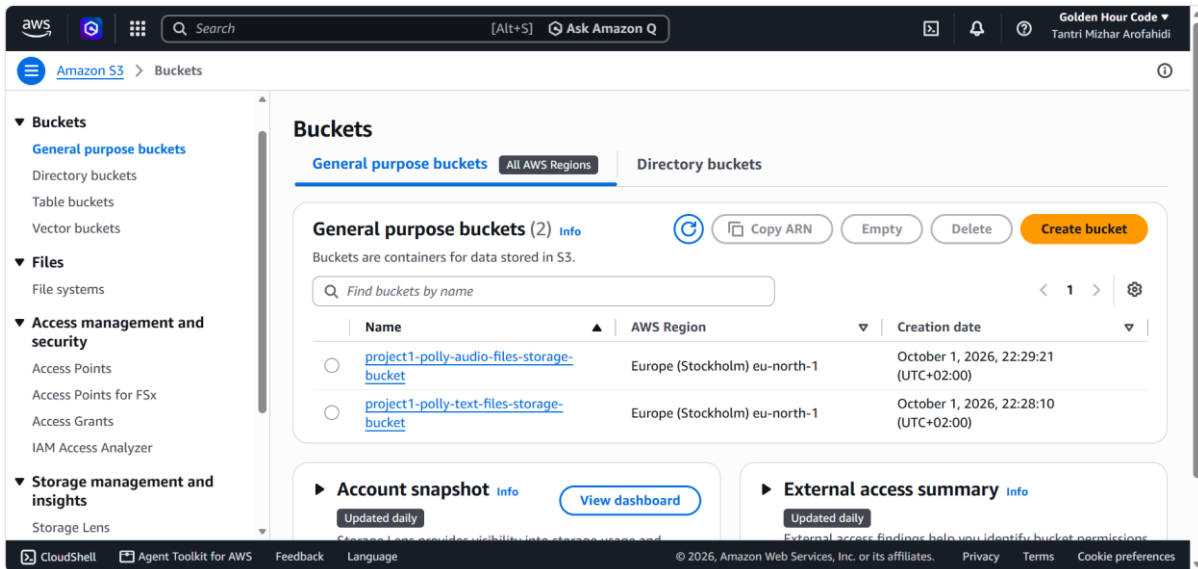
6. Pipeline Design



7. Amazon S3

We would create bucket name in Amazon S3:

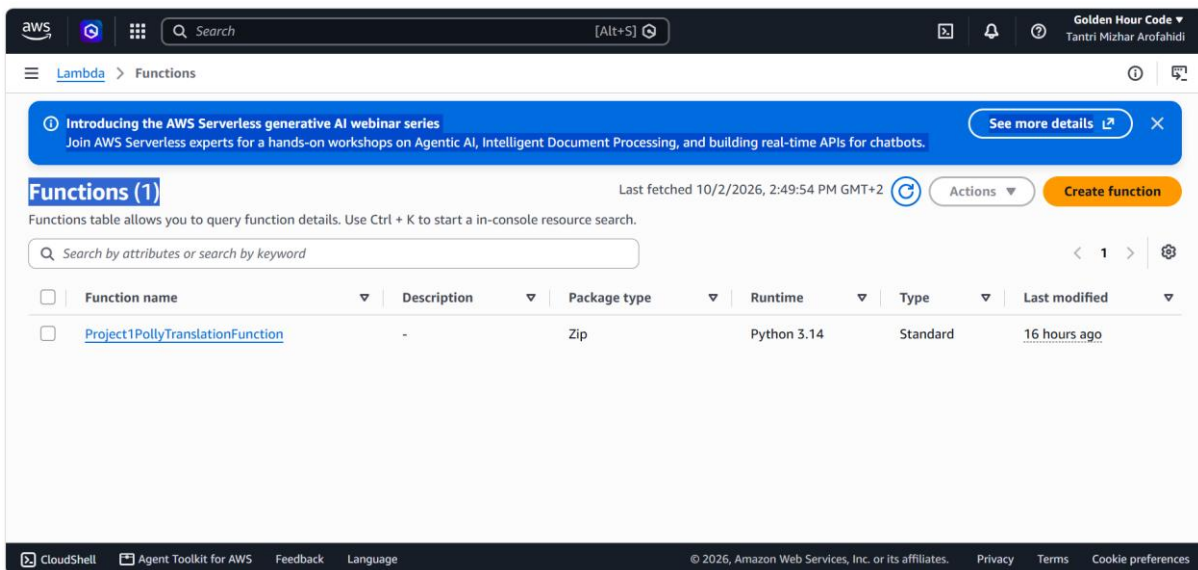
- project1-polly-text-files-storage-bucket
- project1-polly-audio-files-storage-bucket



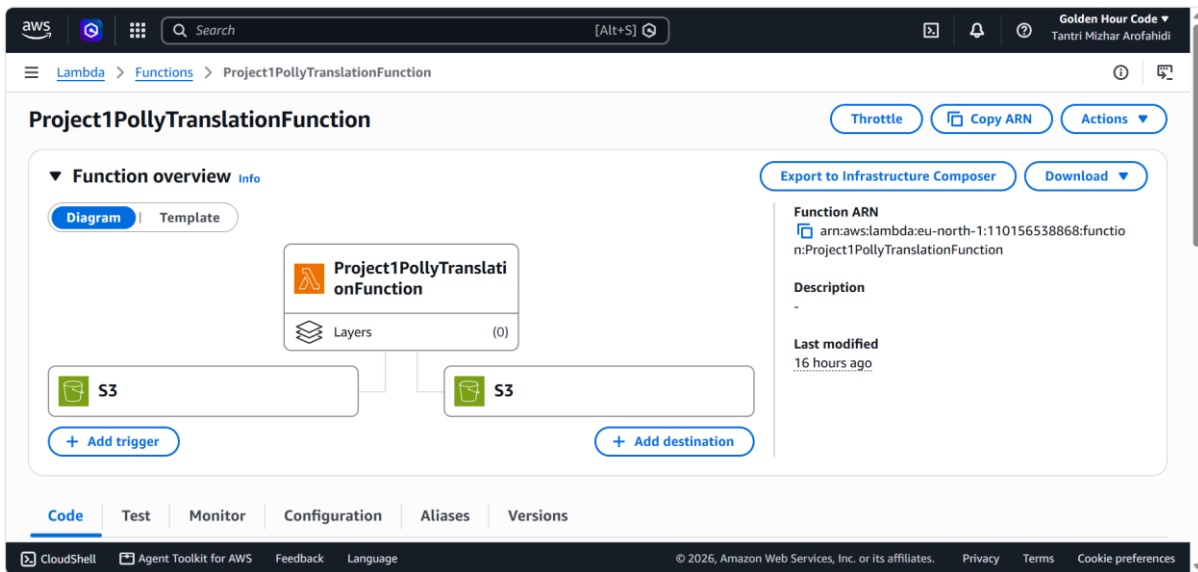
8. Amazon Lambda

a. Configuration amazon Lambda:

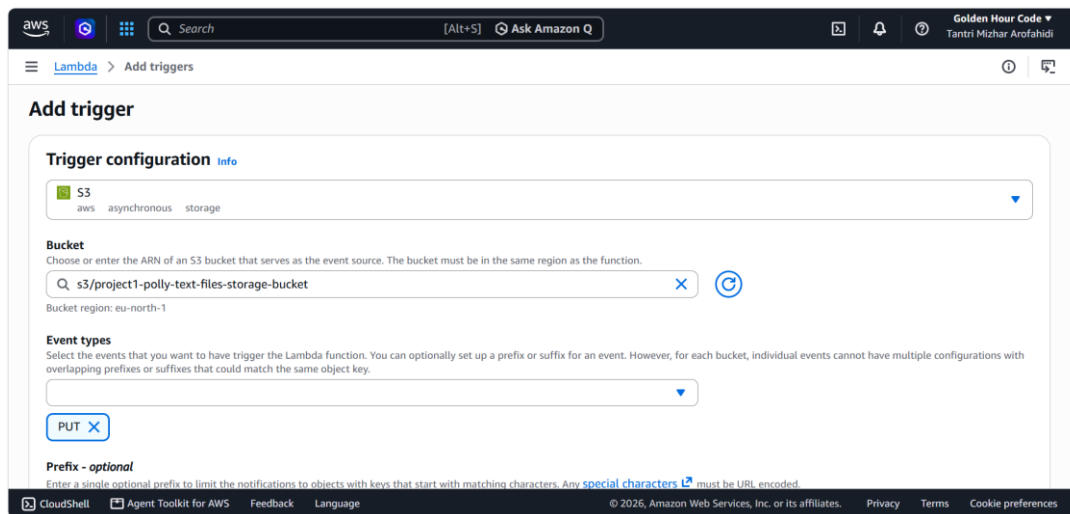
Step 1: Creating name function name “Project1PollyTranslationFunction”



b. Add trigger



We configure Add trigger: Trigger configuration: S3, Bucket: project1-polly-text-files-storage-bucket, event type: PUT, suffix: .txt, Recursive invocation; checklist, add



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Lambda > Add triggers

Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.
e.g. images/

Suffix - optional
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters. Any special characters must be URL encoded.
.txt

Recursive invocation
If your function writes objects to an S3 bucket, ensure that you are using different S3 buckets for input and output. Writing to the same bucket increases the risk of creating a recursive invocation, which can result in increased Lambda usage and increased costs. [Learn more](#)

☒ I acknowledge that using the same S3 bucket for both input and output is not recommended and that this configuration can cause recursive invocations, increased Lambda usage, and increased costs.

Lambda will add the necessary permissions for AWS S3 to invoke your Lambda function from this trigger. [Learn more](#) about the Lambda permissions model.

Cancel Add

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c. Add Destination

We configure Add destination: Source: Asynchronous invocation, Condition: on failure, Destination type: S3, destination: project1-polly-audio-files-storage-bucket, save

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Lambda > Functions > Project1PollyTranslationFunction > Add destination

Add destination

Destination configuration [Info](#)
Configure a destination to receive invocation records. Lambda can send records when your function is invoked asynchronously, or when your function processes records from an event source mapping.

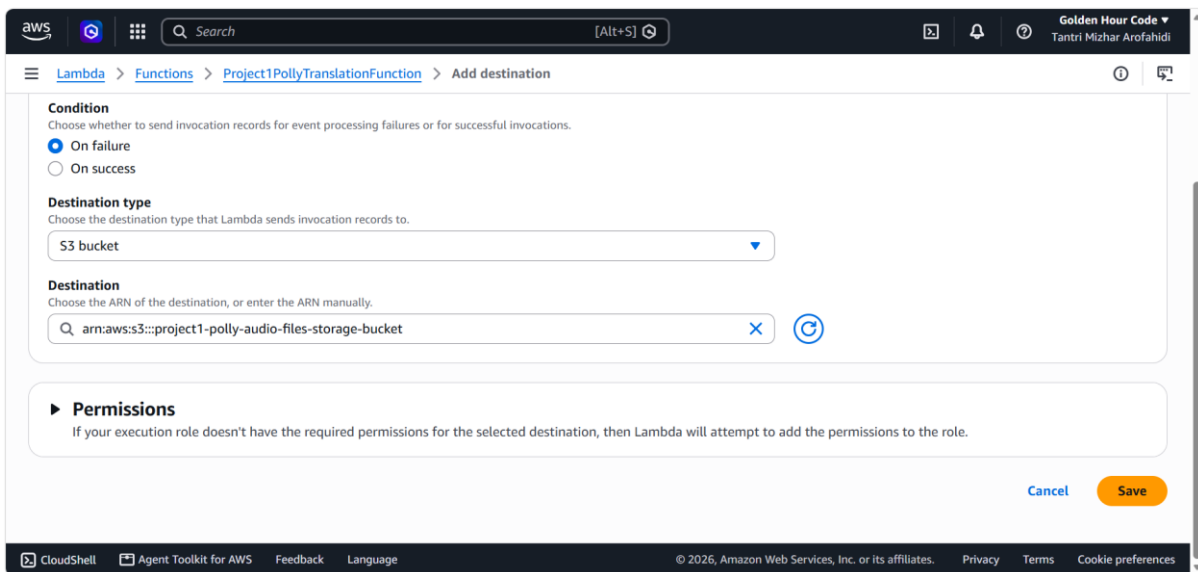
Source
Choose the invocation type that Lambda sends records for.
☒ Asynchronous invocation
☐ Event source mapping invocation

Condition
Choose whether to send invocation records for event processing failures or for successful invocations.
☒ On failure
☐ On success

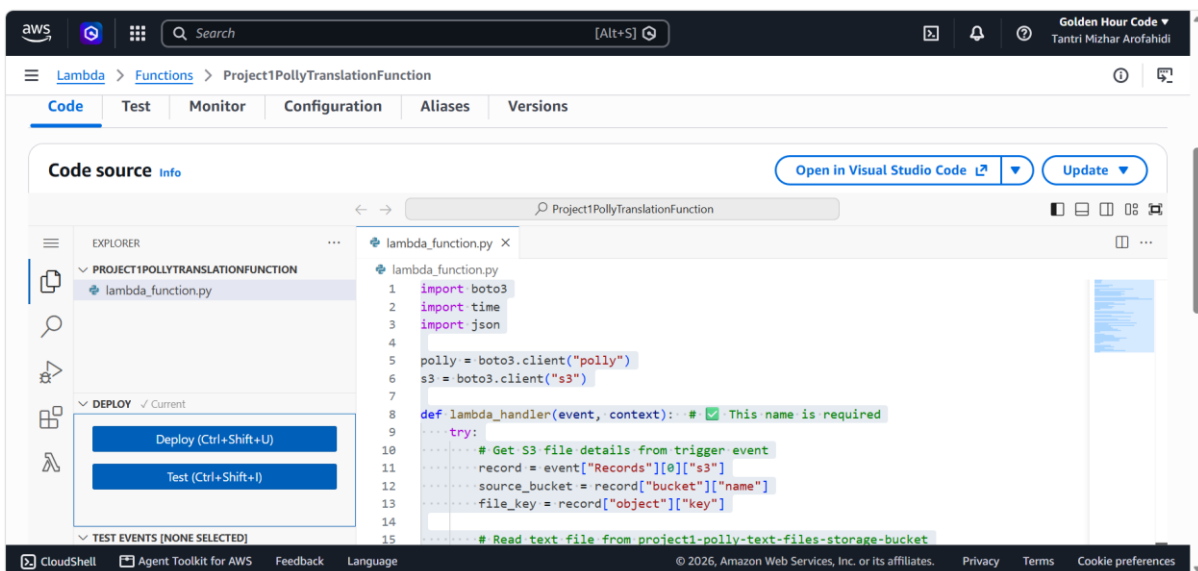
Destination type
Choose the destination type that Lambda sends invocation records to.
S3 bucket

Destination
Choose the ARN of the destination, or enter the ARN manually.

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d. Writing code in Lambda: Copy-paste code in terminal, after that deploy



```
import boto3
```

```
import time
```

```
import json
```

```
polly = boto3.client("polly")
```

```
s3 = boto3.client("s3")
```

```
def lambda_handler(event, context): # ✓ This name is required
```

```
    try:
```

```
        # Get S3 file details from trigger event
```

```

record = event["Records"][0]["s3"]
source_bucket = record["bucket"]["name"]
file_key = record["object"]["key"]

# Read text file from project1-polly-text-files-storage-bucket
obj = s3.get_object(Bucket="project1-polly-text-files-storage-bucket", Key=file_key)
text = obj["Body"].read().decode("utf-8")

# Convert text → speech
polly_response = polly.synthesize_speech(
    Text=text,
    OutputFormat="mp3",
    VoiceId="Joanna"
)

if "AudioStream" not in polly_response:
    raise Exception("Polly did not return audio data")

# Read audio stream
audio_bytes = polly_response["AudioStream"].read()

# Destination audio file name
audio_key = f"speech-{int(time.time()*1000)}.mp3"

# Upload audio to project1-polly-audio-files-storage-bucket
s3.put_object(
    Bucket="project1-polly-audio-files-storage-bucket",
    Key=audio_key,
    Body=audio_bytes,
    ContentType="audio/mpeg",
    ContentLength=len(audio_bytes)
)

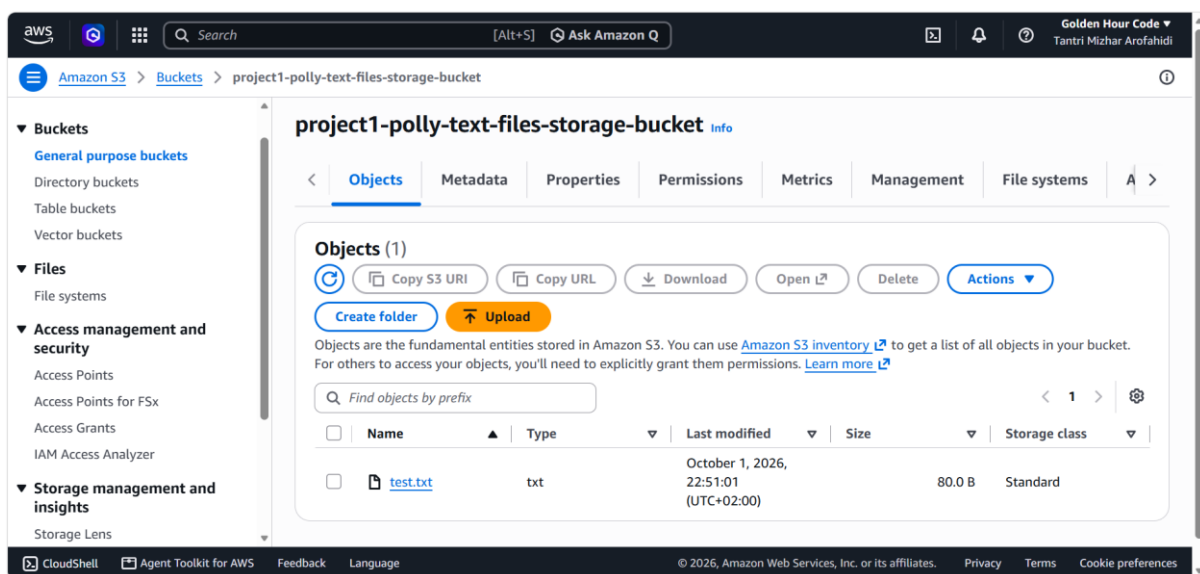
return {
    "statusCode": 200,
    "body": json.dumps({
        "message": "Conversion successful",
        "source_file": file_key,
        "audio_file": audio_key
    })
}

```

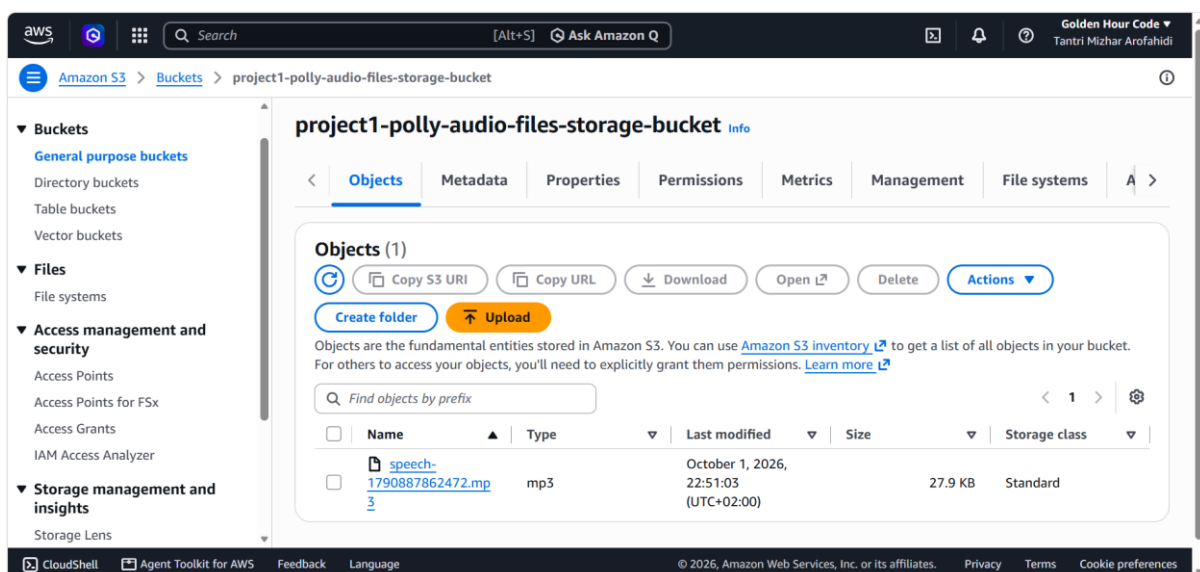
```
except Exception as e:
    print("Error:", str(e))
    return {
        "statusCode": 500,
        "body": json.dumps({"error": str(e)})
    }
```

9. Amazon S3

Upload .txt file (test.txt) after we select “project1-polly-txt-files-storage-bucket”



Then Download file audio which is location in “project1-polly-audio-files-storage-bucket



Finally we gain mp3 audio after we download mp3 after we click “speech-1790887862472.mp3”

The screenshot displays the AWS S3 console interface. The breadcrumb navigation at the top shows the path: Amazon S3 > Buckets > project1-polly-audio-files-storage-bucket > speech-1790887862472.mp3. On the left sidebar, the 'Buckets' section is expanded, showing 'General purpose buckets' as the selected category. The main content area is titled 'speech-1790887862472.mp3' with an 'Info' link. Below the title are three buttons: 'Copy S3 URI', 'Download', and 'Open'. A dropdown menu labeled 'Object actions' is also present. The 'Properties' tab is active, displaying the 'Object overview' section. This section is divided into two columns. The left column contains: Owner (94807f73adb83a78a571dbb5a5de2ccda6e5f786d39bb8799d628f8f6fa5db88), AWS Region (Europe (Stockholm) eu-north-1), Last modified (October 1, 2026, 22:51:03 (UTC+02:00)), and Size (27.9 KB). The right column contains: S3 URI (s3://project1-polly-audio-files-storage-bucket/speech-1790887862472.mp3), Amazon Resource Name (ARN) (arn:aws:s3:::project1-polly-audio-files-storage-bucket/speech-1790887862472.mp3), Entity tag (Etag) (735214c2f09067e82ad6bc8a44a668bc), and Object URL. The bottom of the console shows a dark navigation bar with links for CloudShell, Agent Toolkit for AWS, Feedback, Language, and a copyright notice for Amazon Web Services, Inc. or its affiliates, along with links for Privacy, Terms, and Cookie preferences.

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Amazon S3 > Buckets > project1-polly-audio-files-storage-bucket > speech-1790887862472.mp3

▼ Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets

▼ Files
File systems

▼ Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer

▼ Storage management and insights
Storage Lens

speech-1790887862472.mp3 Info

Copy S3 URI Download Open Object actions

Properties Permissions Versions

Object overview

Owner 94807f73adb83a78a571dbb5a5de2ccda6e5f786d39bb8799d628f8f6fa5db88	S3 URI s3://project1-polly-audio-files-storage-bucket/speech-1790887862472.mp3
AWS Region Europe (Stockholm) eu-north-1	Amazon Resource Name (ARN) arn:aws:s3:::project1-polly-audio-files-storage-bucket/speech-1790887862472.mp3
Last modified October 1, 2026, 22:51:03 (UTC+02:00)	Entity tag (Etag) 735214c2f09067e82ad6bc8a44a668bc
Size 27.9 KB	Object URL

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